ASSIGNMENT-16.4

TASK 1: Database Schema Design

Schema SQL

```sql
CREATE TABLE Instructors (
    instructor_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100),
    email VARCHAR(100),
    country VARCHAR(50)
);

CREATE TABLE Courses (
    course_id INT PRIMARY KEY AUTO_INCREMENT,
    course_name VARCHAR(100),
    instructor_id INT,
    FOREIGN KEY (instructor_id) REFERENCES Instructors(instructor_id)
);

CREATE TABLE Students (
    student_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100),
    email VARCHAR(100)
);

CREATE TABLE Enrollments (
    enrollment_id INT PRIMARY KEY AUTO_INCREMENT,
    student_id INT,
    course_id INT,
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);
```

TASK 2: Populate Database

Query SQL

```sql
INSERT INTO Instructors (name, email, country) VALUES
('John', 'john@gmail.com', 'USA'),
('Ravi', 'ravi@gmail.com', 'India'),
('Sara', 'sara@gmail.com', 'UK');

INSERT INTO Courses (course_name, instructor_id) VALUES
('AI Basics', 1),
('DBMS', 2),
('Python', 2),
('ML', 1);

INSERT INTO Students (name, email) VALUES
('Amit', 'amit@gmail.com'),
('Priya', 'priya@gmail.com'),
('Kiran', 'kiran@gmail.com');

INSERT INTO Enrollments (student_id, course_id) VALUES
(1,1),
(2,1),
(3,2),
(1,3);
```

OUTPUT:-

Results

Query #1 Execution time: 15.11ms

There are no results to be displayed.

Query #2 Execution time: 1.78ms

There are no results to be displayed.

Query #3 Execution time: 1.17ms

There are no results to be displayed.

Query #4 Execution time: 1.43ms

There are no results to be displayed.

Results

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Query #6 Execution time: 0.11ms

course_id	course_name	instructor_id
1	AI Basics	1
2	DBMS	2
3	Python	2
4	ML	1

Query #7 Execution time: 0.13ms

student_id	name	email
1	Amit	amit@gmail.com
2	Priya	priya@gmail.com
3	Kiran	kiran@gmail.com

Query #8 Execution time: 0.09ms

enrollment_id	student_id	course_id
1	1	1
2	2	1
3	3	2
4	1	3

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TASK 3: CRUD Operations

```sql
UPDATE Students
SET email = 'amit_new@gmail.com'
WHERE student_id = 1;

SELECT * FROM Students;

DELETE FROM Courses
WHERE course_id = 4;


SELECT * FROM Courses;
```

OUTPUT:-

Query #9 Execution time: 0.53ms

There are no results to be displayed.

Query #10 Execution time: 0.11ms

student_id	name	email
1	Amit	amit_new@gmail.com
2	Priya	priya@gmail.com
3	Kiran	kiran@gmail.com

Query #11 Execution time: 0.52ms

There are no results to be displayed.

Query #12 Execution time: 0.16ms

course_id	course_name	instructor_id
1	AI Basics	1
2	DBMS	2
3	Python	2

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TASK 4: Aggregate Queries

```sql
SELECT course_id, COUNT(student_id) AS total_students
FROM Enrollments
GROUP BY course_id;
```

OUTPUT:-

Query #13 Execution time: 0.2ms

course_id	total_students
1	2
2	1
3	1

TASK 5: Query Optimization

```sql
SELECT c.course_name
FROM Courses c
JOIN Instructors i
ON c.instructor_id = i.instructor_id
WHERE i.country = 'India';

CREATE INDEX idx_country ON
Instructors(country);
```

OUTPUT:-

3	1

Query #14 Execution time: 0.17ms

course_name
DBMS
Python

Query #15 Execution time: 22.6ms

There are no results to be displayed.